

AI Search Framework v2: Executive Presentation Brief

Headline Finding: An all-SLM pipeline ($\leq 8B$) with feedback loop delivers **41.6% cost savings over 70B LLMs** and **88.9% savings over Frontier APIs**, while cutting graph decomposition errors by **58.3%**.

1. Parameter Scale Cost Benchmark (N=80)

Baseline System	Parameter Scale	Cost Ratio (SLM / Baseline)	Cost Savings (%)	95% Confidence Interval
Llama-3.1-8B	8.0B	2.656×	-165.6% (Overhead)	[2.508, 2.805]
Qwen-2.5-32B	32.5B	1.178×	-17.8%	[1.112, 1.244]
Llama-3.1-70B	70.6B	0.584×	+41.6% Savings	[0.551, 0.616]
Qwen-2.5-72B	72.7B	0.584×	+41.6% Savings	[0.551, 0.616]
Gemini-1.5-Pro	Frontier	0.111×	+88.9% Savings	[0.104, 0.117]

2. Independent Blind Judge Win-Rates (Claude-3.5-Sonnet)

- 🥇 Qwen-2.5-72B: 25.0% (20 wins)
- 🥈 Gemini-1.5-Pro: 18.75% (15 wins)
- 🥉 Llama-3.1-70B: 15.0% (12 wins)
- 🥉 Llama-3.1-8B: 15.0% (12 wins)
- 💎 SLM Pipeline v2: 13.75% (11 wins)
- 💎 Qwen-2.5-32B: 12.5% (10 wins)

3. Feedback Loop Structural Accuracy Gain

- Graph Edit Distance:** Reduced from **3.57** → **1.50**.
- Relative Error Reduction:** **58.33%** structural accuracy improvement.
- Economic Crossover:** Located at $\approx$ **35B parameters**.